RYAN BRODNAX

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SUMMARY

Dynamic and results-driven **Senior Data Engineer** with proven expertise in **collaborative team environments** and delivering **exceptional data engineering solutions** within the **Microsoft Azure ecosystem**. Adept at leveraging **Azure Data Factory**, **Azure Databricks**, **Azure Synapse Analytics**, and **SQL** to architect and implement scalable, secure, and automated data pipelines and solutions. Excels in **cross-functional collaboration**, mentoring, and aligning data strategies with business objectives to drive measurable outcomes. Committed to **continuous learning**, adaptability, and thriving in fast-paced environments with a focus on **delivery excellence** and client satisfaction.

SKILLS

- **Programming Languages:** Python, SQL, T-SQL, Scala, C#, Java, JavaScript, PowerShell, R, JSON
- **Frameworks & Libraries:** Apache Spark, Pandas, NumPy, TensorFlow, Keras, XGBoost, LightGBM, Dynamics 365, Power Platform, .NET
- **Tools & Platforms:** Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Azure SQL Database, Azure Data Lake Storage (ADLS), Azure DevOps, Visual Studio, Git, OAuth 2.0, REST APIs, Databricks Lakehouse Platform, Databricks, Power BI, Microsoft Azure Portal, Jenkins
- **Databases:** Azure SQL Database, SQL Server, NoSQL, PostgreSQL, MySQL, Oracle Database, Azure Cosmos DB, Cassandra, Redis, SQLite
- **DevOps & CI/CD:** Azure DevOps, Jenkins, Git, GitHub, JIRA, Terraform, Docker, Kubernetes, Ansible, Apache Airflow
- **Other Skills:** Data Governance, Data Orchestration, Metadata Driven Pipelines, Dimensional Modeling, Star Schemas, Data Security, Machine Learning, Model Deployment, API Integration, Data Migration, Real-time Analytics, Problem Solving, Client Collaboration, Project Management

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CERTIFICATIONS

- Microsoft Certified: Azure Data Engineer Associate
- Microsoft Certified: Azure Solutions Architect Expert
- Microsoft Certified: Azure Fundamentals

EXPERIENCE

Senior Data Engineer (Microsoft Azure)

CVS Health • Feb 2020 - Present

- Led AI-driven data personalization projects leveraging **Azure Databricks**, **Azure Data Factory**, and **SQL** to enhance medication adherence analytics across 10,000+ retail locations, improving patient outcomes by 1.6%.
- Architected scalable data processing pipelines with **Azure Data Lake Storage (ADLS)**, **Azure Databricks**, and **Python**, enabling efficient transformation and integration of pharmacy and biometric datasets.
- Developed automated ETL workflows using **Azure Data Factory**, **SQL**, and **Azure Synapse Analytics** to streamline ingestion from multiple data sources while ensuring enterprise-grade data security and compliance.
- Collaborated cross-functionally with business intelligence teams to design dimensional models and star schemas in **Azure SQL Database** facilitating advanced reporting and predictive analytics.
- Engineered Python-based machine learning models using collaborative filtering and predictive analytics frameworks deployed on **Databricks Lakehouse Platform** to improve patient outreach and adherence programs.
- Implemented OAuth 2.0 secured REST APIs to integrate external health data sources with **Azure Databricks** and **Azure Data Factory**, enhancing real-time data availability for clinical insights.
- Mentored junior data engineers on best practices involving **Azure Databricks** usage, **data pipeline orchestration**, and advanced **SQL** query optimization strategies.
- Spearheaded data migration initiatives from on-premise systems to **Azure Synapse Analytics** ensuring zero downtime data availability and maintaining stringent data governance policies.
- Optimized Spark jobs running on **Azure Databricks** clusters by fine-tuning configurations and leveraging **Scala** and **Python** libraries for improved latency and scalability.
- Developed metadata-driven pipelines in **Azure Data Factory** minimizing manual interventions and accelerating data integration cycle times across multiple projects.
- Integrated data security principles combining **data lake** and **data warehouse** strategies to ensure HIPAA compliance while maintaining high performance in analytics workloads.
- Led technical workshops on Azure data services including **Azure Data Factory**, **Databricks**, and **Synapse Analytics** ensuring team-wide technical excellence and adoption of modern data engineering practices.
- Enforced data quality and reliability standards across pipelines using unit testing frameworks and custom validation scripts in **Python** and **SQL**.
- Collaborated with cloud administrators to configure secure access policies in **Azure Data Lake Storage** and manage performance-scaling for critical pipelines.

- Designed and developed notification and alerting mechanisms within **Azure Data Factory** to monitor pipeline health and failures proactively.
- Applied modular design principles in data engineering solution architecture to facilitate reusability and reduce technical debt across projects.
- Enabled real-time analytics integration by configuring event-based triggers in **Azure Data Factory** and streaming ingestion into **Azure Synapse Analytics**.
- Automated deployment of infrastructure using **Azure DevOps** pipelines and **Terraform** scripts ensuring consistent environment management.
- Utilized advanced SQL and T-SQL scripting to join, aggregate, and transform complex relational datasets spanning transactional and historical data sources.
- Championed a culture of **continuous learning** and knowledge sharing via regular code reviews and internal technical webinars focused on **Azure cloud data services**.

Senior ML Engineer

Wells Fargo • Apr 2017 - Feb 2020

- Designed and implemented enterprise-scale data pipelines utilizing **Azure Data Factory**, **Azure Databricks**, and **Azure SQL Database** to support real-time fraud detection and financial risk analytics systems.
- Developed feature engineering workflows on **Apache Spark** clusters hosted on **Azure Databricks** for over 1,400 machine learning models deployed to production with **SQL** data sources.
- Employed advanced **SQL** queries and **dimensional modeling** techniques within **Azure Synapse Analytics** to optimize data warehouse schemas for risk analytics and capital markets applications.
- Integrated external API data sources embedded with **OAuth 2.0** authentication protocols into staging environments using **Azure Data Factory** pipelines for enriched data sets.
- Led design sessions with cross-functional teams and client stakeholders applying Microsoft Azure best practices in **data orchestration** and modernization strategies.
- Implemented metadata-driven orchestration mechanisms enabling dynamic pipeline adjustments according to evolving business requirements and high-volume event streams.
- Architected technical governance frameworks around secure data lakehouse principles blending **Azure Data Lake Storage** with **Azure Synapse Analytics** for regulation-compliant data handling.
- Optimized batch and streaming data workflows ensuring latency below 100 milliseconds using **Databricks** runtime optimizations and Spark Structured Streaming.
- Collaborated with infrastructure teams deploying CI/CD pipelines in **Azure DevOps** automating unit tests, builds, and release cycles for data engineering applications.
- Facilitated hands-on training and mentorship programs on **Azure Databricks**, **SQL** optimization, and metadata-driven pipeline architecture best practices.
- Developed RESTful APIs for ML model inference embedded within the **Azure ecosystem**, ensuring secure, scalable access to prediction services.
- Authored extensive documentation and architectural diagrams championing maintainability, reuse, and clarity across the data engineering lifecycle.

- Guided cross-team agile ceremonies improving communication and ensuring delivery timelines aligned with project goals and quality standards.
- Conducted performance tuning on **SQL Server** and **Azure SQL Database** backends to handle high concurrency and data volume for risk analytic dashboards.
- Deployed and configured Azure Synapse Analytics workspaces with modular scripts focusing on automation and repeatability across projects.
- Applied best practices in **data security**, auditing, and access control policies using role-based access and encryption within **Microsoft Azure**.
- Automated anomaly detection pipelines leveraging **Python** and **ML libraries** integrated with Azure data platforms to provide early warning of suspicious activities.
- Integrated GPU-accelerated machine learning using NVIDIA technology via containerized environments orchestrated with **Kubernetes** and Azure Kubernetes Service.
- Managed large-scale data ingestion workflows combining batch and streaming ingestion from on-prem and cloud sources to **Azure Data Lake Storage**.
- Advocated and practiced ownership and accountability by leading critical incident responses and delivering robust solutions to complex data engineering challenges.

Machine Learning Engineer

Sway AI • Aug 2016 - Apr 2017

- Built performant convolutional neural network (CNN) models for image classification, segmentation, and detection using **TensorFlow**, **Keras**, and **Python** on GPU-enabled cloud infrastructure.
- Developed large-scale data pipelines to preprocess and ingest image datasets into storage solutions with high throughput using **Azure Data Lake Storage** and **Azure Databricks**.
- Implemented model optimization techniques including pruning and quantization improving inference speed and reducing deployment costs on **Azure cloud** platforms.
- Created and deployed scalable RESTful APIs for ML model inference leveraging **Azure Functions** and **Azure API Management**, ensuring secure, reliable access.
- Collaborated with cross-functional teams applying **source control tools** like **Git** integrated with Azure DevOps pipelines for smooth continuous integration and delivery cycles.
- Automated data validation and augmentation processes within **Azure Databricks** notebooks using **Python** libraries to enhance training data quality and model robustness.
- Conducted performance profiling and debugging of deep learning workflows to optimize utilization of **Azure GPU** resources and reduce job completion time.
- Implemented OAuth and JSON flattening techniques for secure data ingestion and transformation pipelines feeding production image analytics models.
- Participated in architectural design discussions for future AI product features focusing on scalable, maintainable data engineering and ML solutions in the cloud.
- Enforced modularity and code reuse principles by establishing reusable components and templates for ML lifecycle management using Azure tools.

- Delivered comprehensive technical documentation and training sessions to onboard peers onto Microsoft cloud AI solutions and pipeline best practices.
- Applied modern security practices ensuring compliance with data privacy policies in machine learning model training and deployment processes.
- Led troubleshooting efforts utilizing logs, telemetry, and monitoring tools native to Azure cloud environments for rapid root cause analysis.
- Designed experiments and proof-of-concept projects to evaluate emerging deep learning frameworks and new Azure services relevant to computer vision.
- Created robust workflows combining **Python**, **TensorFlow** pipelines, and **Azure Databricks** orchestrations tailored for continuous model improvement cycles.
- Coordinated with DevOps teams to automate infrastructure provisioning using **Terraform** and containerized workloads via **Docker** on **Azure Kubernetes Service**.
- Monitored training data pipelines and implemented failover mechanisms leveraging Azure event-driven architectures for enhanced system resiliency.
- Engaged in full data science lifecycle including feature engineering, model validation, and deployment within Microsoft Azure powered environments.
- Established key performance indicators and metrics dashboards using **Power BI** reflecting operational data insights for AI deployment effectiveness.
- Encouraged a culture of best practices and continuous feedback in data engineering through knowledge sharing and adherence to Microsoft recommended patterns.

EDUCATION

Master's Degree in Computer Science - 2016

Alliance University

Bachelor's Degree in Computer Science - 2014

Alliance University